Project 1: An Analysis of Genshin Impact

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Introduction

Genshin Impact is a popular role-playing game developed and published by miHoYo, which is a Chinese game development company. The game was released globally on September 28, 2020, and has received widespread attention from players who are keen on role-playing games around the world, even people who never get in touch with this kind of game are indulgent in it. The company and this special game have been praised for a long time because of the unique systems and special operations needed in Genshin Impact. Some main features bring it into people's views. First, as designed around the concept of an open-world environment, it allows people to freely climb mountains, swim in the water, and run in different kinds of areas, and players can even interact with the apples and special fruit on the tree, flowers and butterflies that can be picked up, and old wooden barrels and dried firewoods that can be broken and burnt (Figure 1). Here comes the second feature of the game, which is the elemental combat system. There are a total of seven elements in the game, that are Anemo, Electro, Hydro, Pyro, Cryo, Dendro, and Geo. They can interact with each other in all sorts of ways, which is the primary allure of the system. From the example of burning firewood that we mentioned earlier, it can be observed that combustion reactions can happen among Dendro and Pyro. What's more, the beautiful visuals and soothing soundtrack attract a lot of people to stay in the game without doing anything, just enjoying the beautiful surroundings. The technical and art team in miHoYo aims to develop an appearance system to let people from different countries admire the world without the need to go abroad. For example, a country called Liyue in the game is designed with inspiration from various Chinese cultures, landforms, and geographic environments. Players hanging around in Liyue in the game can have a relatively specific understanding of how the terrains and cultures in China feel.



Figure 1. Burning Firewoods

Usability Principles

Strive for Consistency

This principle emphasizes consistency when users interact with the system. The system should maintain the same level of functioning models and technical terms, as well as the design of interaction buttons and word appearance should look similar and have a harmonious outlook using similar elements. The main idea of a functioning principle of the system should not be changed. This helps users create an expectation of what would happen if they select a similar interaction button as before. It requires different dialog boxes and other interface elements to use similar layouts, controls, and terminology to be consistent.

In Genshin Impact, when players log into the game, the background is very similar to the story scene when the game starts (Figure 2). People analyze that every time we log into the game, it refers to the main character entering the world. It is a consistency shown in a very detailed point. The story in the game is rather important. It built a stage that made everything reasonable. Also, the elemental reactions are used throughout the whole game. The game encourages players to combine different elemental abilities to trigger powerful reactions that can be easily understood. Striving for consistency in applying elemental reactions during combat is crucial for optimizing damage and strategic advantages.

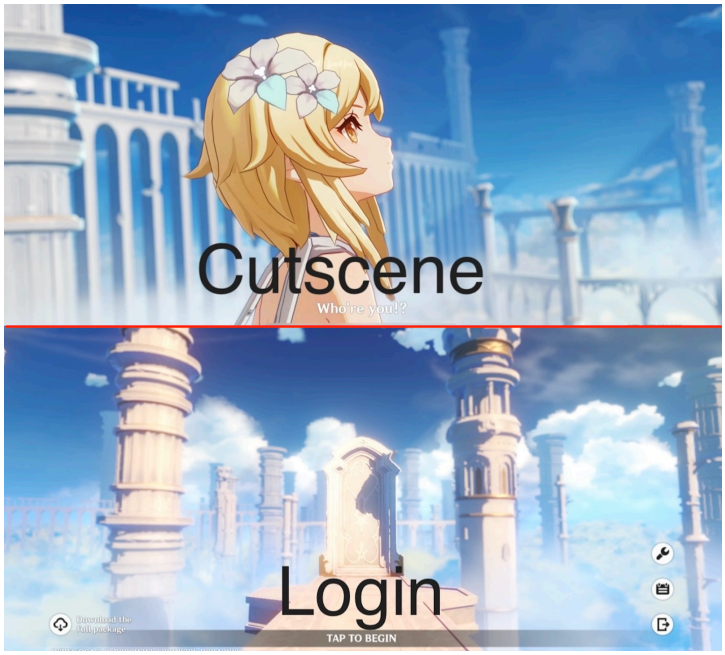


Figure 2. The login page is similar to the cutscene shown at the game starts.



Figure 3. The monster gets frozen by an interaction of Hydro and Cryo.

It is similar as we put a cup of water into a refrigerator, the water will freeze (Figure 3). There are many other reactions among any two elements. For example, Cryo and Pyro will end up with melt. We can imagine that snow melts when the weather gets warm. Observing the reaction between different elements in the game is also an interesting part since it is also consistent with the real world. Also, playing Genshin Impact can improve the understanding of chemical reactions to some extent.

Seek universal usability.

This principle aims to allow a wide range of users to interact with the system and also allows professional users to quickly visit some areas. One system only facing several small groups of people shows that it is weak in universal usability.

Genshin Impact meets the needs of different user groups, including users of different ages, cultures, ability levels, and usage backgrounds. First of all, it allows players to set their language preferences. There are a total of 15 kinds of game languages and 4 kinds of voice-over languages that can be selected (Figure 4). It is a respect to different language speakers. What's more, the beginner's instructions and follow-up navigations are detailed and specific. It allows people above 12 years of age to have a rough understanding of how to interact with the system and enjoy the game. All the navigations are saved in "Tutorials" (Figure 5). Also, the compatibility of Genshin Impact with various gaming platforms, including Windows, PlayStation, iOS, Android, and macOS, ensures that players can enjoy the game seamlessly across a

wide range of devices (Figure 6). Besides, Genshin Impact allows players to custom shortcut keys, which is convenient, especially for Joystick players.

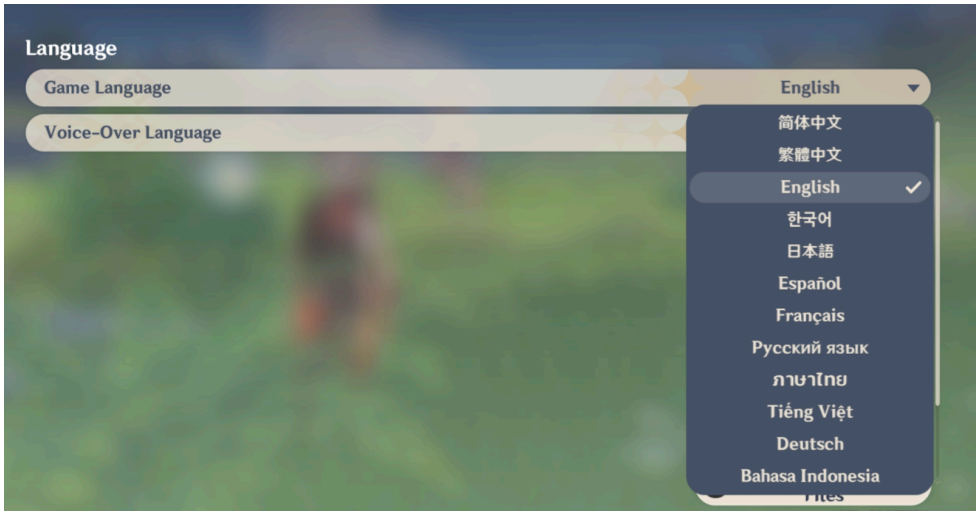


Figure 4. List of available languages including Chinese, English, and Japanese.

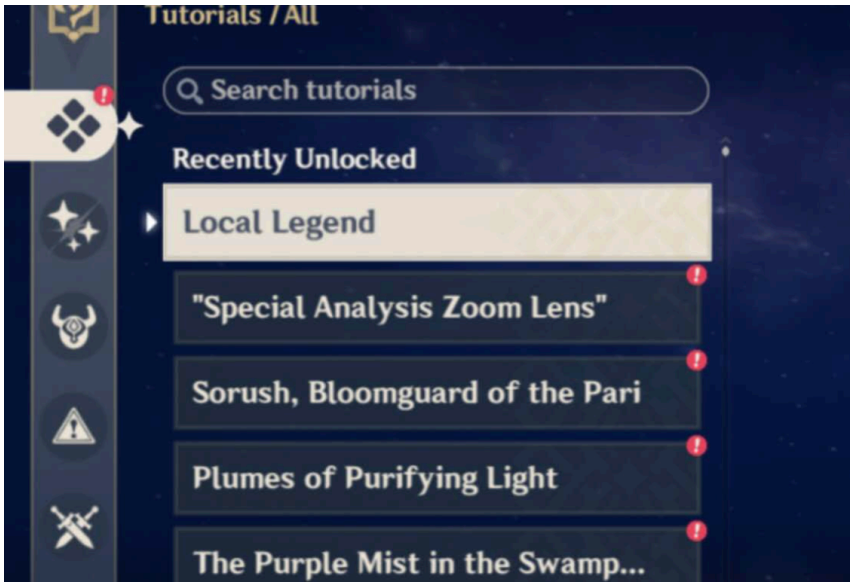


Figure 5. "Tutorials" will show the recently unlocked guides first.

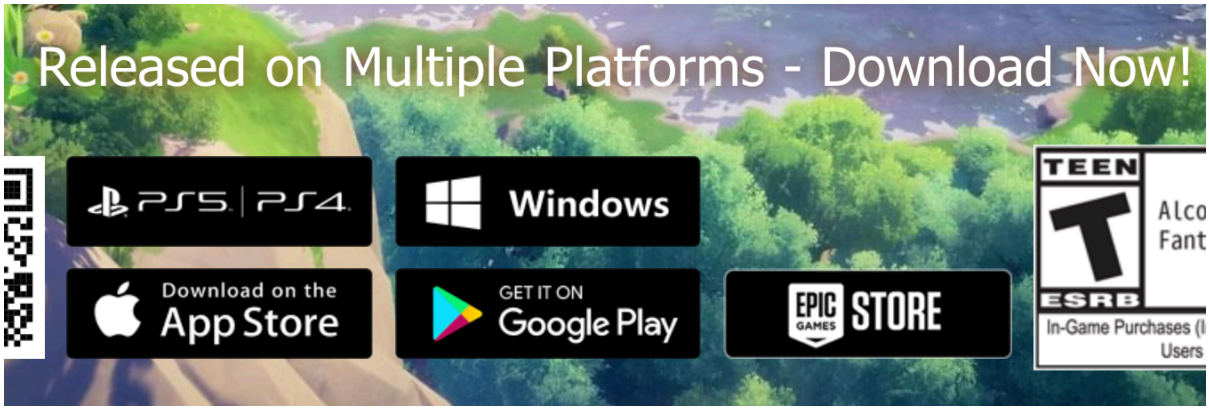


Figure 6. Genshin Impact is released on multiple platforms, including PS5, PS4, Windows, iOS

Offer informative feedback

When users interact with the system, there should be feedback in time to clearly show the result of the interaction or the status of the system to users. The informative feedback can help users understand what has happened by reviewing the feedback. When a user acts, the system should pop up a message to provide feedback to confirm that the action has been completed successfully. Otherwise, there should be an error alert showing that the user's action causes an error, at the same time, a guidance message should be displayed to solve the error.

It is important for a game to offer informative feedback. In Genshin Impact, when there comes an elemental reaction, the name of the interaction will appear near where it happened. In Figure 3, the word "Freeze" provides players with the information that there is a reaction of frozen. At the same time, some numbers are coming out. The numbers represent the damage to an item when characters deal or receive damage. This immediate and quantifiable feedback allows players to assess the effectiveness of their attacks and understand the impact of enemy actions. What's more, Genshin Impact also displays the load status to the players to let them know that the system is working and not unresponsive (Figure 7).



Figure 7. Load status is designed with the icons of 7 Elements.

Design dialogs to yield closure

This principle emphasizes that every dialog box or interface should be designed logically with a clear beginning and end, guide the users step by step, and give clear feedback showing that the series of actions and guides are done. It allows users to have a better understanding of how to complete by doing each step.

There is a reward system in Genshin Impact that can exactly show how many quests are done. Players can click “Claim Rewards”, and a pop-up will show a list of rewards they just received. After you claim the rewards, a green check mark is shown. And a forbidden mark is shown for the rewards that you can no longer get (Figure 8). Players are passionate about completing quests, since “informative feedback at the completion of a group of actions gives users the satisfaction of accomplishment, a sense of relief” (Shneiderman, Ben, et al, 2018, *Designing the User Interface: Strategies for Effective Human-Computer Interaction*, 6th edition). Completing the quests not only gives players resources that can be used in the game but also provides them with achievements.



Figure 8. List of daily commission rewards.

Accessibility Principles

Perceivable

Perceivable refers to providing information and content in a way that is accessible and understandable by seeing or touching. Genshin Impact allows players to set graphics, for example, players can adjust brightness (Figure 9), and you can see the feedback immediately when dragging the button to adjust. Also, the texts are well arranged with suitable font sizes, and players can adjust the volumes of music, dialogue, and SFX separately. When a character is speaking, the “audio with speech has no or very low background noise so the speech is easily distinguished”. (WebAIM) The character’s words can be heard clearly, and players are allowed to better appreciate the work of the voiceover artists.

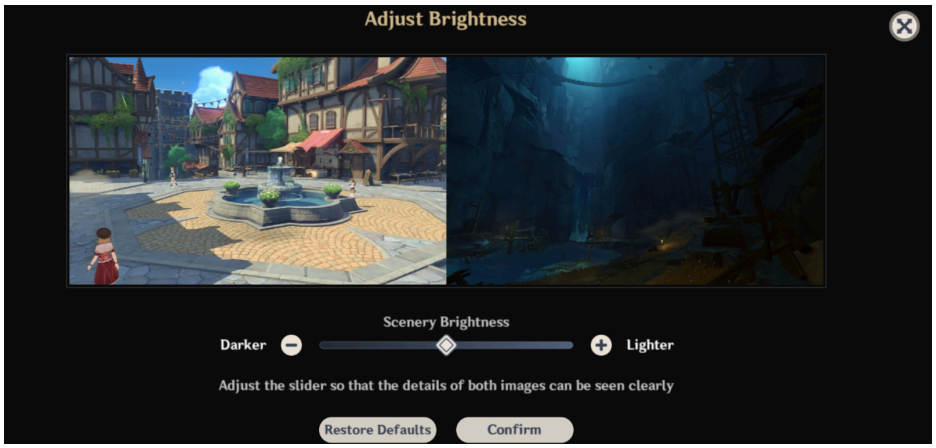


Figure 9. Adjust Brightness

Operable

A system is operable when it is compatible with multiple electric equipment such as a mouse, a keyboard, and so on. Genshin Impact allows players to choose their control types such as touchscreen in the Controls page in Settings (Figure 10). Most of the players prefer to play Genshin Impact on their computers with only keyboards and mice. Some of the users “can navigate to and from all navigable page elements using only a keyboard.” (WebAIM) Also, as the official mentioned, it is recommended to play with headphones since the soundtracks are amazing and almost every player agrees with that.

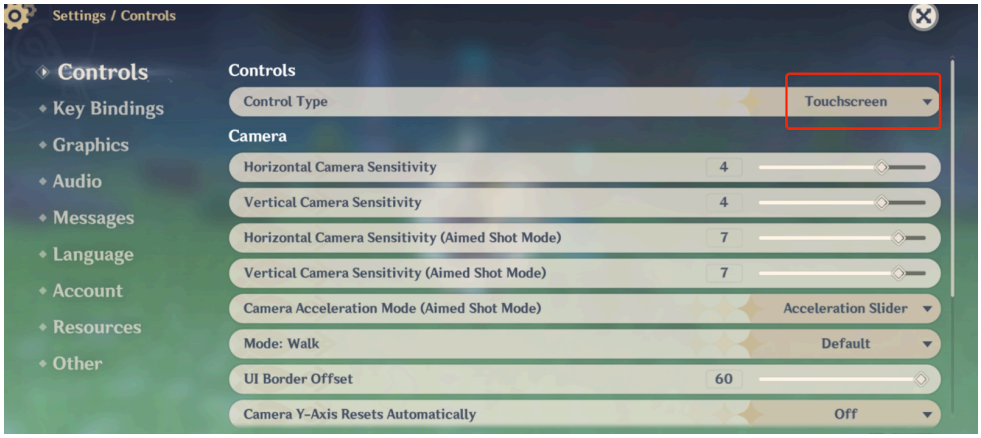


Figure 10. the Controls page in Settings

Extra Analysis: Language System

For consistency, Genshin Impact created some new languages in the game. They are sorted as common scripts, scripts in different countries, and the language that monsters use. The common script is used in a wide range, especially in tutorial images. In a tutorial, a page of instructions is given, which uses this common script that players cannot read. However, the script is consistent and can be recognized by analyzing the similar scripts that appeared in the game. It can be treated as the universal script across the world in the game, differentiated from the language that appears in each country.

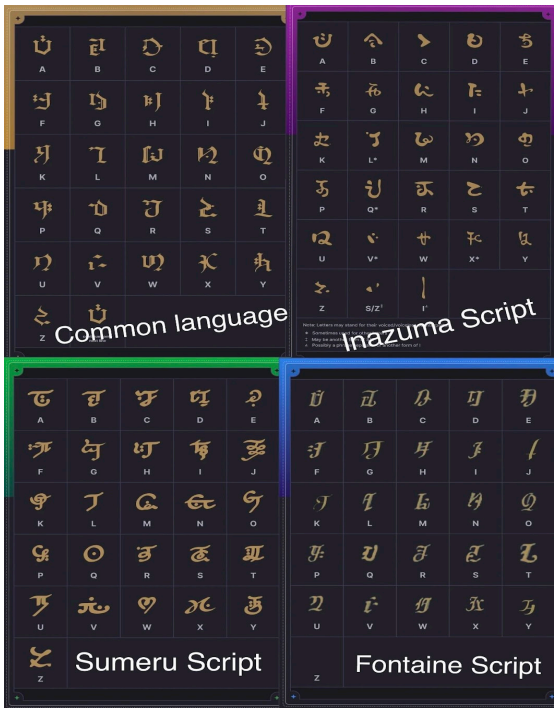


Figure 11. Scripts of some languages in the game compared with alphabets in the real world

When players are located in different countries in the game, they can find that there are some special scripts for the specific country and culture. Enthusiastic players have come up with some comparison tables about scripts shown up in the game and alphabets in the real world (Figure 11). Even though the scripts are in disparate languages and environments, the appearance of each letter representing the same alpha letter is pretty similar. Also, insisting on using unreadable scripts on tutorials has improved the consistency of the writing system. Through vision, the design and rules of the writing system immerse players in a lifelike experience. What's more, players are allowed to find more interesting details that are not shown in the main storyline. That is also the necessary part to make this game so fascinating. Particularly, for the monster language, there is a special voice-in system since no scripts can be recognized by monsters. There is even an activity to allow players to learn how to communicate with monsters under a non-playable character's instruction, who is a scholar of monsters' linguistics in the game (Figure 12).



Figure 12. The scholar called Ella is trying to say hi to the awake monster.

Works Cited

Shneiderman, Ben, et al. “3.3.4 The Eight Golden Rules of Interface Design.” *Designing the User Interface: Strategies for Effective Human-Computer Interaction*, 6th ed., Pearson, Harlow, Essex, 2018.

WebAIM, “WebAIM's WCAG 2 Checklist - Perceivable.” Institute for Disability Research, Policy, and Practice, 12 Oct. 2023. <WebAIM's WCAG 2 Checklist - Perceivable>

WebAIM, “WebAIM's WCAG 2 Checklist - Operable.” Institute for Disability Research, Policy, and Practice, 12 Oct. 2023. <WebAIM's WCAG 2 Checklist - Operable>

Appendix

When there comes the text of a kind of language, it symbolizes cultural progress. In Genshin Impact, I found that the monsters with low cultural literacy do not have their script, while in the society of humans and gods, there exist many kinds of scripts. I searched online and found a conclusion of the comparison to the English character, also got a screenshot of the scholar who has tried so many times talking with a monster in the game. In terms of appearance and quantity, the various languages and scripts designed in this game have a certain degree of uniformity and consistency. Also, studying the meaning of words helps us to gain insight into the nature of communication both for humans and for monsters. In real life, learning about languages can also reveal the cultural, cognitive, and social factors behind language. Research in this field contributes to a better understanding of the complexity of language and promotes deeper thinking about linguistic phenomena, and it applies to games as well. Since the monsters can speak, the game mentions that they were humans in the past. That's why they are able to communicate in a particular language. Official recommends players think about the linguistic phenomena among monsters, which is related to the main story of the game, and is related to the closure of the game.